Linear algebra qualification exam

May 4, 2026

1. In this and the next problem, $\mathcal{P}_m(F)$ denotes the vector space of all polynomials of degree at most m over the field F.

   Define $T \in \mathcal{L}(\mathcal{P}_m(\mathbb{R}))$ by the rule $T(p) = p' + p(0)x^m$.

   (a) Write the matrix for T according to the basis $\{1, x, x^2, \ldots, x^m\}$.

   (b) Show that $T^{m+1} = (m!)I$.

2. Prove that there exist a unique set of numbers $a, b, c \in \mathbb{R}$ such that for all quadratic polynomials $p \in \mathcal{P}_2(\mathbb{R})$, $p(4) = ap(1) + bp(2) + cp(3)$.

3. Let $v_1, \ldots, v_n$ be a basis of the vector space V, and $v_1^*, \ldots, v_n^* \in V^*$ be the dual basis. Prove that the map $V \to V$ defined by

$$x \mapsto \sum_{i=1}^{n} v_i^*(x) v_i$$

   is the identity.

4. Suppose T is a normal operator on a finite-dimensional complex inner product space V.

   (a) Show that T and T^* (the adjoint of T) have a common eigenvector v.

   (b) Show that if $\langle v, w \rangle = 0$ for some $w \in V$, then $\langle v, Tw \rangle = 0$.

5. This is a continuation of Problem 4 above. Show that T is diagonalizable with respect to an orthonormal eigenbasis. You can use 4(a) and 4(b) even if you did not prove them.